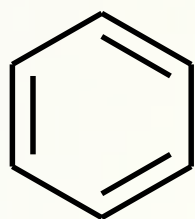


Chương 7: Arene

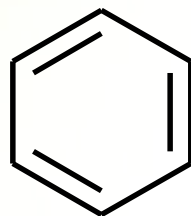
I. CẤU TẠO BENZEN

1. Cấu tạo benzen

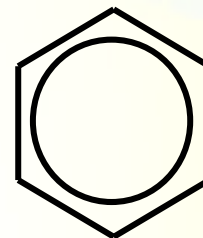
Benzen có CTPT C_6H_6



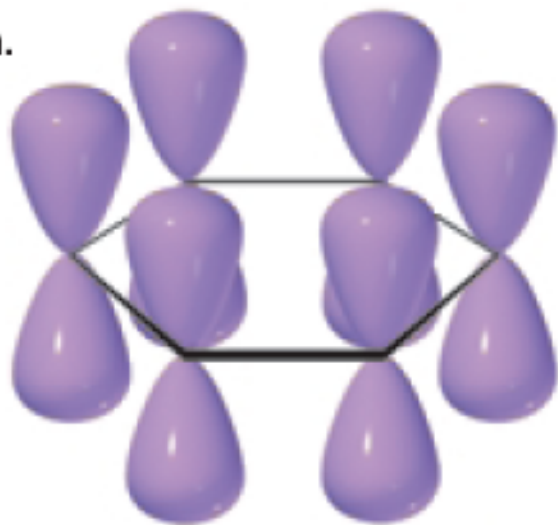
hoặc



$\Rightarrow$



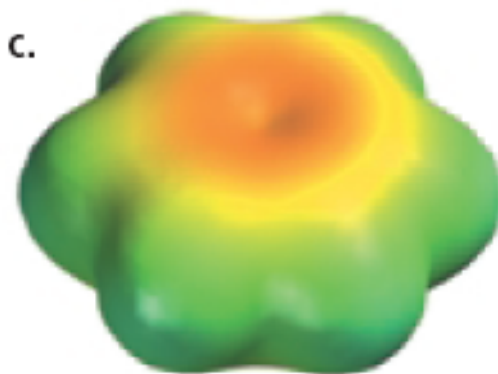
a.



b.



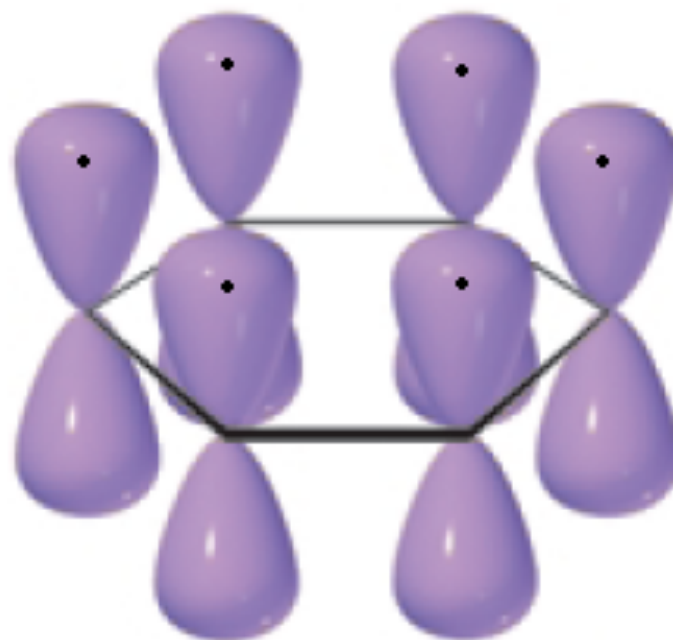
c.



I. CẤU TẠO BENZEN

2. Tính thơm

Quy tắc Hukel: Hydrocacbon thơm (aren) – là những hợp chất vòng liên hợp có cấu tạo phẳng với số điện tử p bằng $4n + 2$, $n = 0, 1, 2, \dots$



I. CẤU TẠO BENZEN

2. Tính thơm



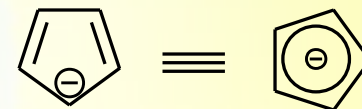
cyclopropene cation cyclopropenil



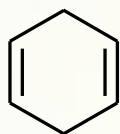
cyclobuta-1,3-diene



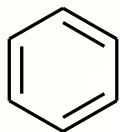
cyclopenta-1,3-diene



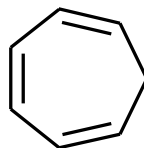
anion cyclopentadienil



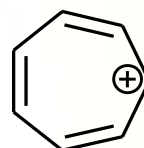
cyclohexa-1,4-diene



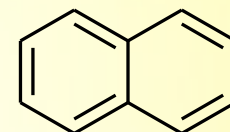
benzene



cyclohepta-1,3,5-triene

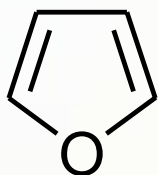


cation cycloheptatrienil

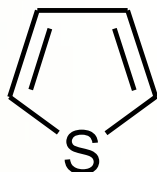


naphthalene

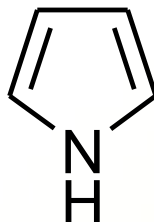
Một số hợp chất dị vòng có tính thơm:



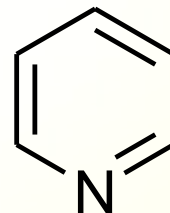
furan



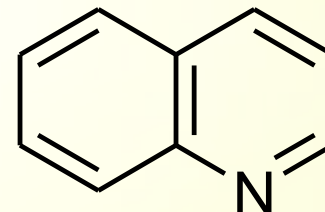
thiophene



pyrrole



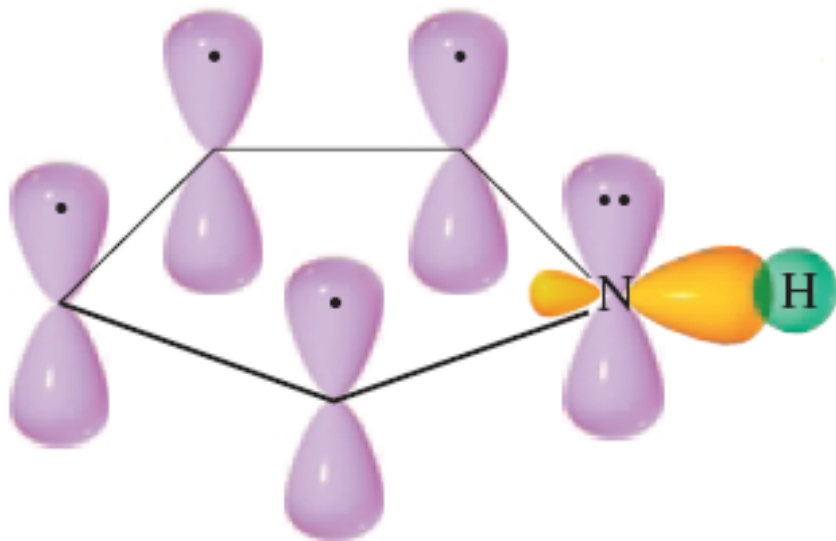
pyridine



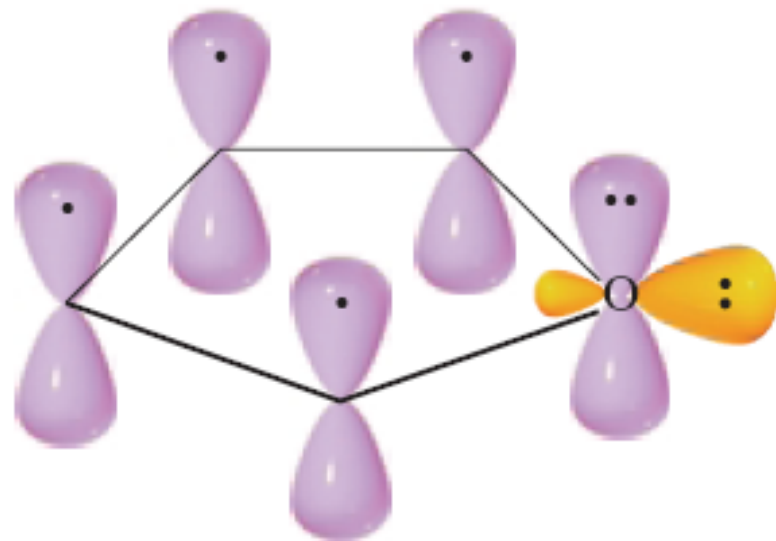
quinoline

I. CẤU TẠO BENZEN

2. Tính thơm



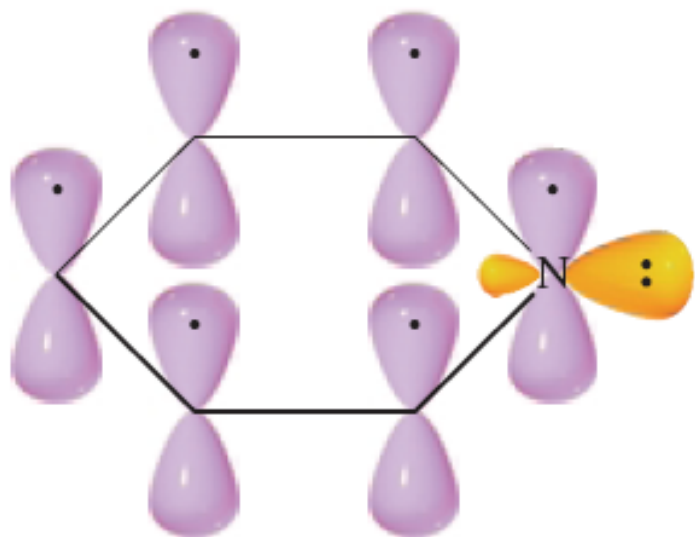
orbital structure of pyrrole



orbital structure of furan

I. CẤU TẠO BENZEN

2. Tính thơm

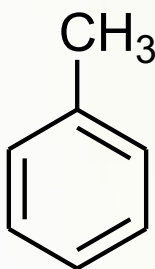


orbital structure of pyridine

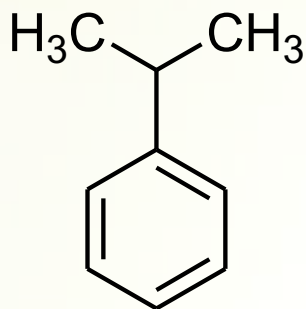


II. DANH PHÁP

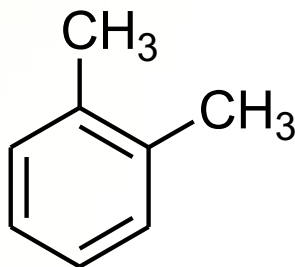
1. Tên thông thường



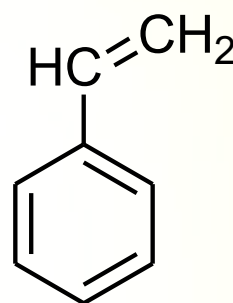
toluene



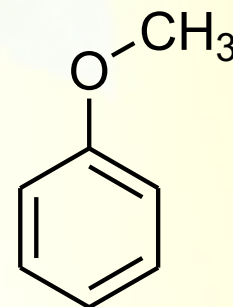
cumene



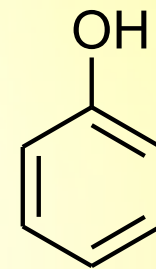
o-xylene



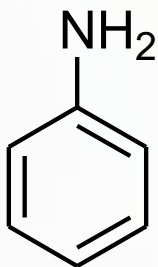
styrene



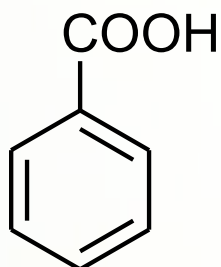
anisole



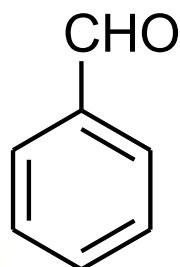
phenol



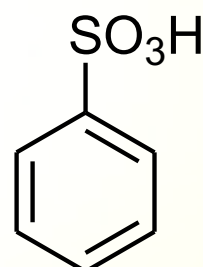
aniline



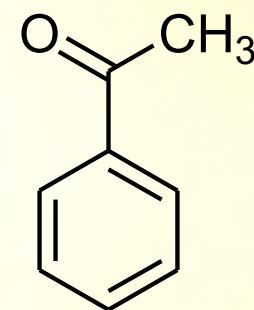
benzoic acid



benzaldehyde



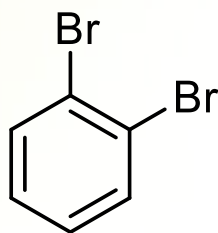
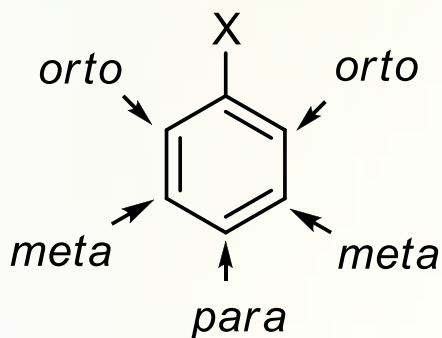
benzenesulfonic acid



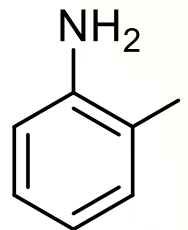
acetophenone

II. DANH PHÁP

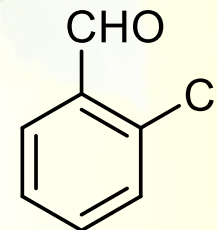
2. Tên IUPAC



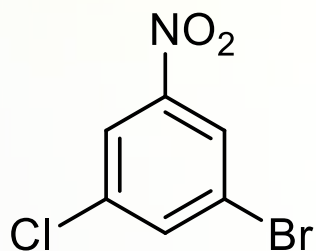
o-dibromobenzene



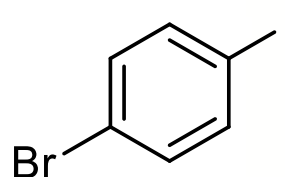
o-iodoanilin



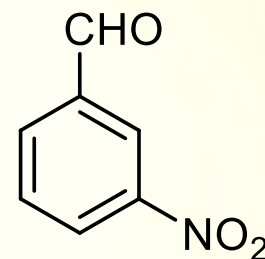
2-chlorobenzaldehyde



1-bromo-3-chloro-5-nitrobenzene



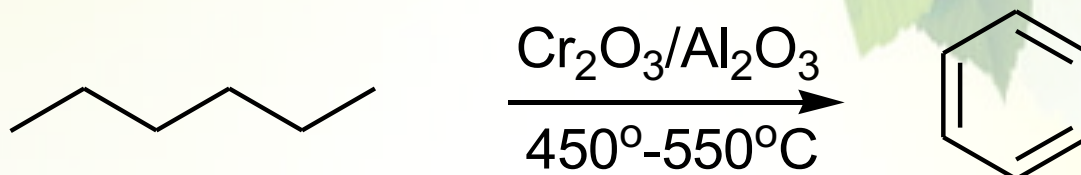
1-bromo-4-iodobenzene
p-bromoiodobenzene



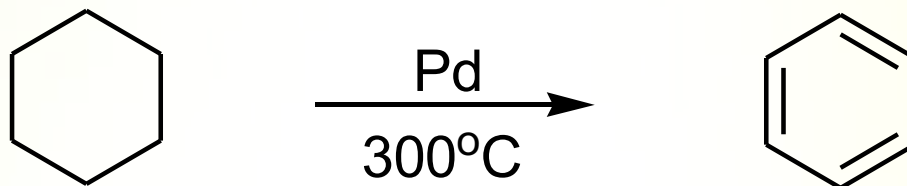
3-nitrobenzaldehyde

III. ĐIỀU CHẾ

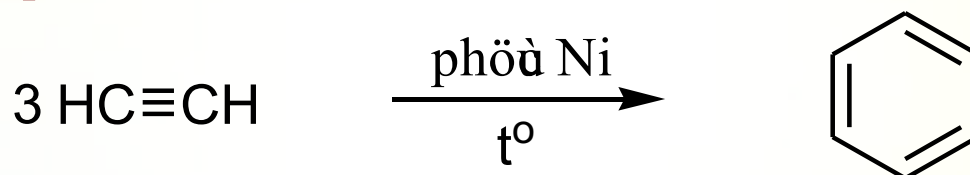
1. Đóng vòng và dehydro hoá ankan



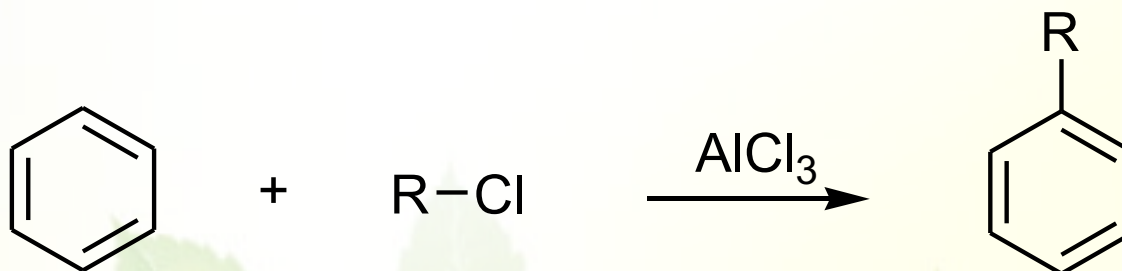
2. Dehydro hoá cycloankan



3. Từ acetylen



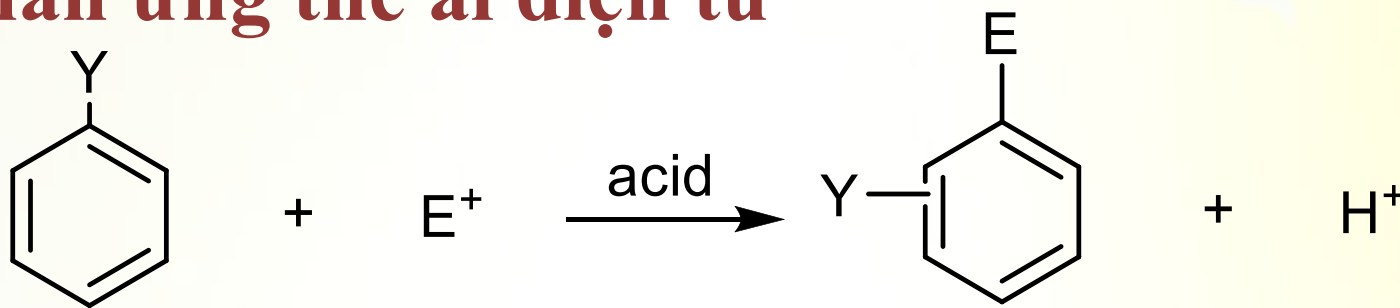
4. Ankyli hoá benzen theo Friedel-Crafts



V. TÍNH CHẤT HOÁ HỌC

Không cho phản ứng cộng hợp, cho phản ứng thế.

• Phản ứng thế ái điện tử



XT: H_2SO_4 , H_3PO_4 hay axit Levis AlCl_3 , FeCl_3 , ZnCl_2 ..

Khả năng thế: Y là nhóm thế sẵn có trên vòng benzene

Nếu : Y – là nhóm hút điện tử, giảm khả năng phản ứng

Y – là nhóm đẩy điện tử, tăng khả năng phản ứng

V. TÍNH CHẤT HOÁ HỌC

Quy luật thế:

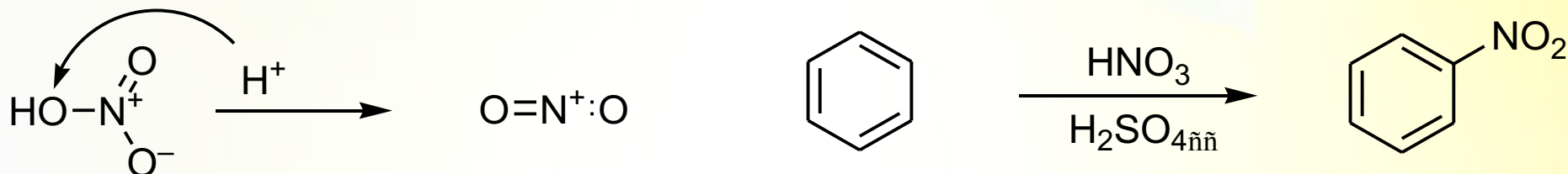
Y – là nhóm hút điện tử (-NO, -NO₂, -CN, -CHO, -COOH, -COOR, CONHR), → E⁺ thế vào vị trí *meta* so với Y

Y- là nhóm đẩy điện tử (-R, -OR, -NH₂, -OH, -SH, -OOCR, -NHOCR), → E⁺ thế vào vị trí *ortho* và *para* so với Y

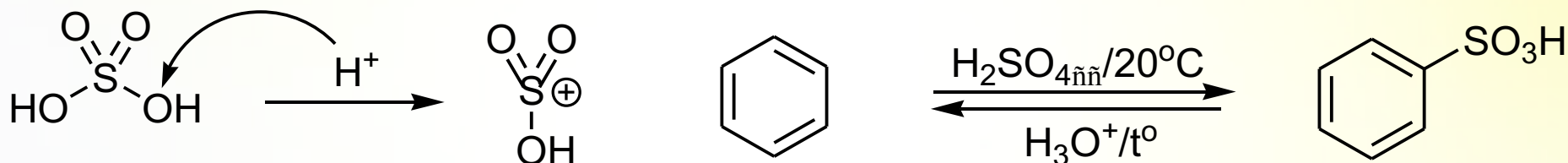
V. TÍNH CHẤT HOÁ HỌC

2. Các phản ứng thế ái điện tử tiêu biểu

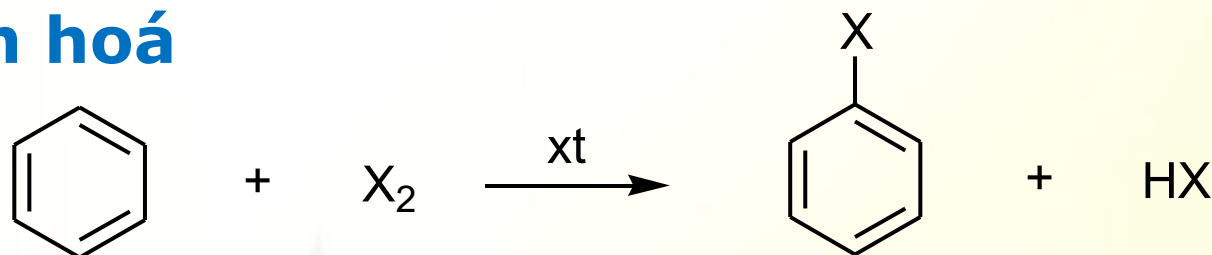
a. Nitro hoá



b. Sunfo hoá



c. Halogen hoá

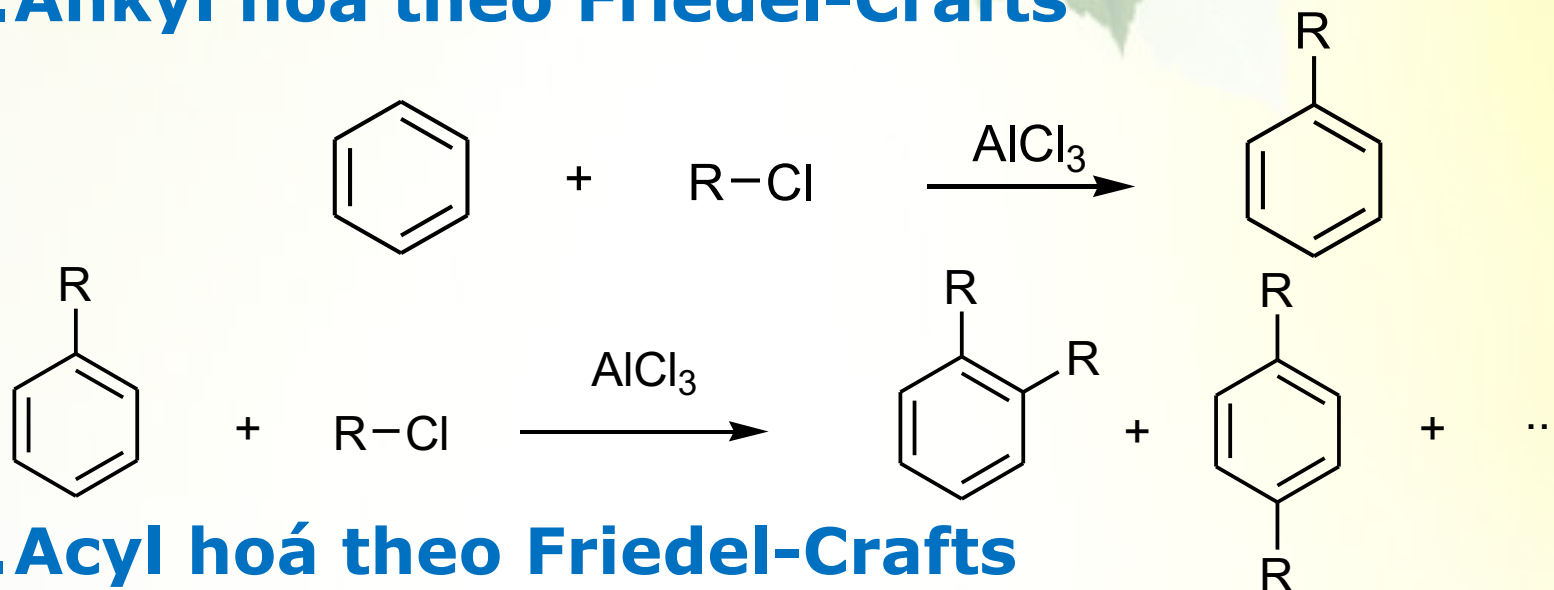


xuất xứ: axit Lewis AlCl_3 , FeCl_3 , ZnCl_2 ...

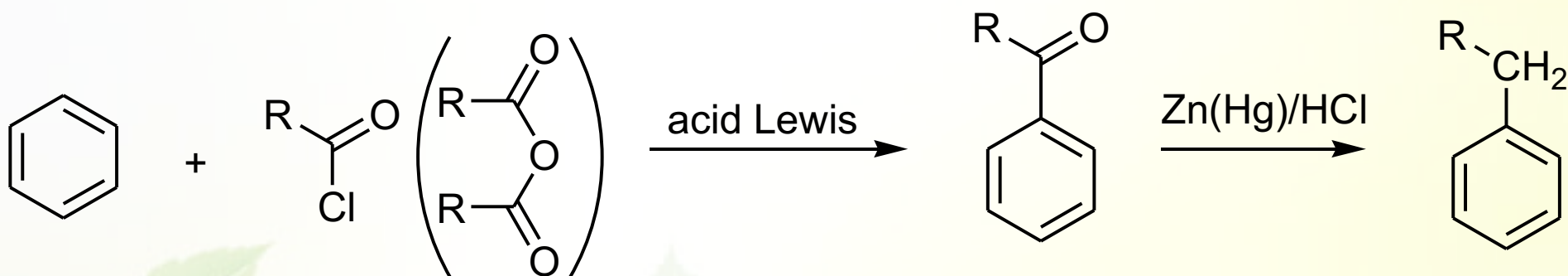
V. TÍNH CHẤT HOÁ HỌC

2. Các phản ứng thế ái điện tử tiêu biểu

d. Ankylation according to Friedel-Crafts



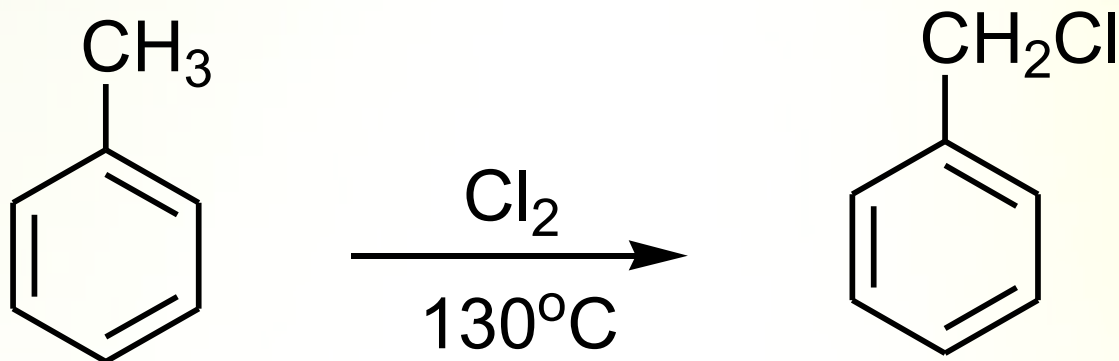
e. Acylation according to Friedel-Crafts



V. TÍNH CHẤT HOÁ HỌC

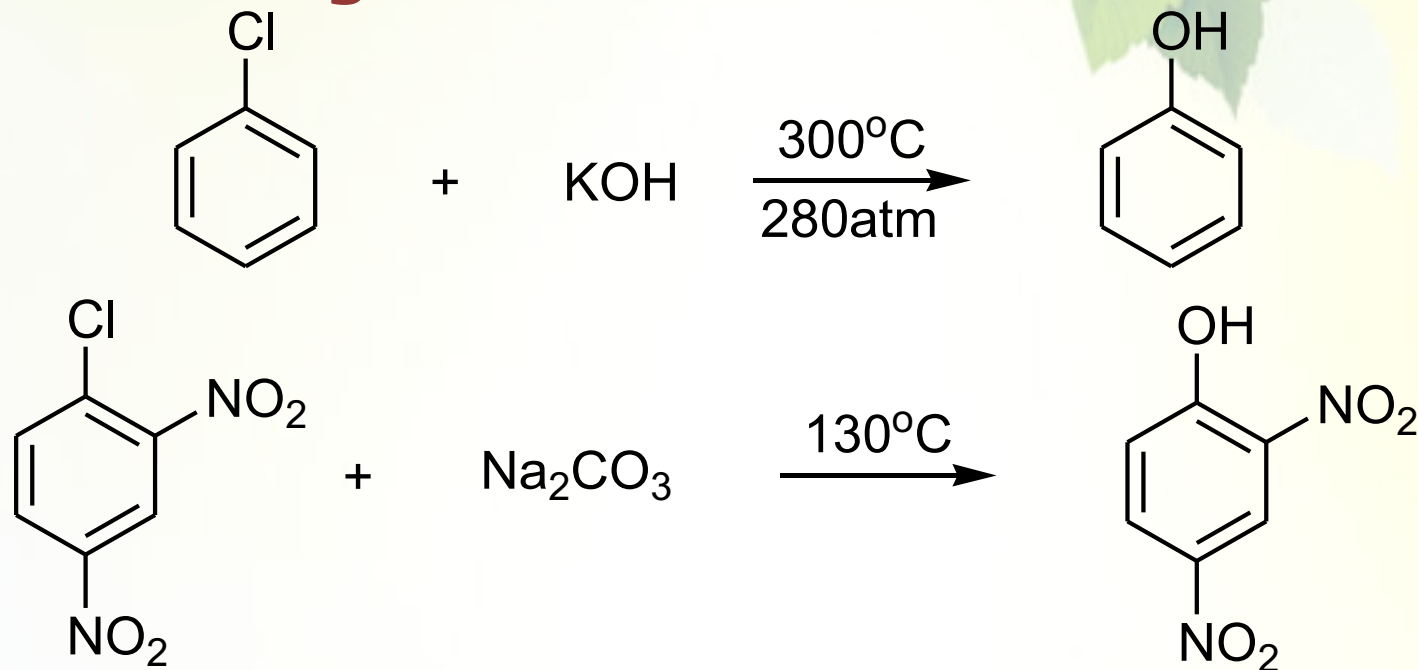
2. Các phản ứng thế ái điện tử tiêu biểu

Halogen hoá mạch nhánh: (S_R)



V. TÍNH CHẤT HOÁ HỌC

3. Phản ứng thế ái nhân



4. Phản ứng oxi hoá

